

Invertebrates

AN INVERTEBRATE IS AN ANIMAL WITHOUT A BACKBONE.

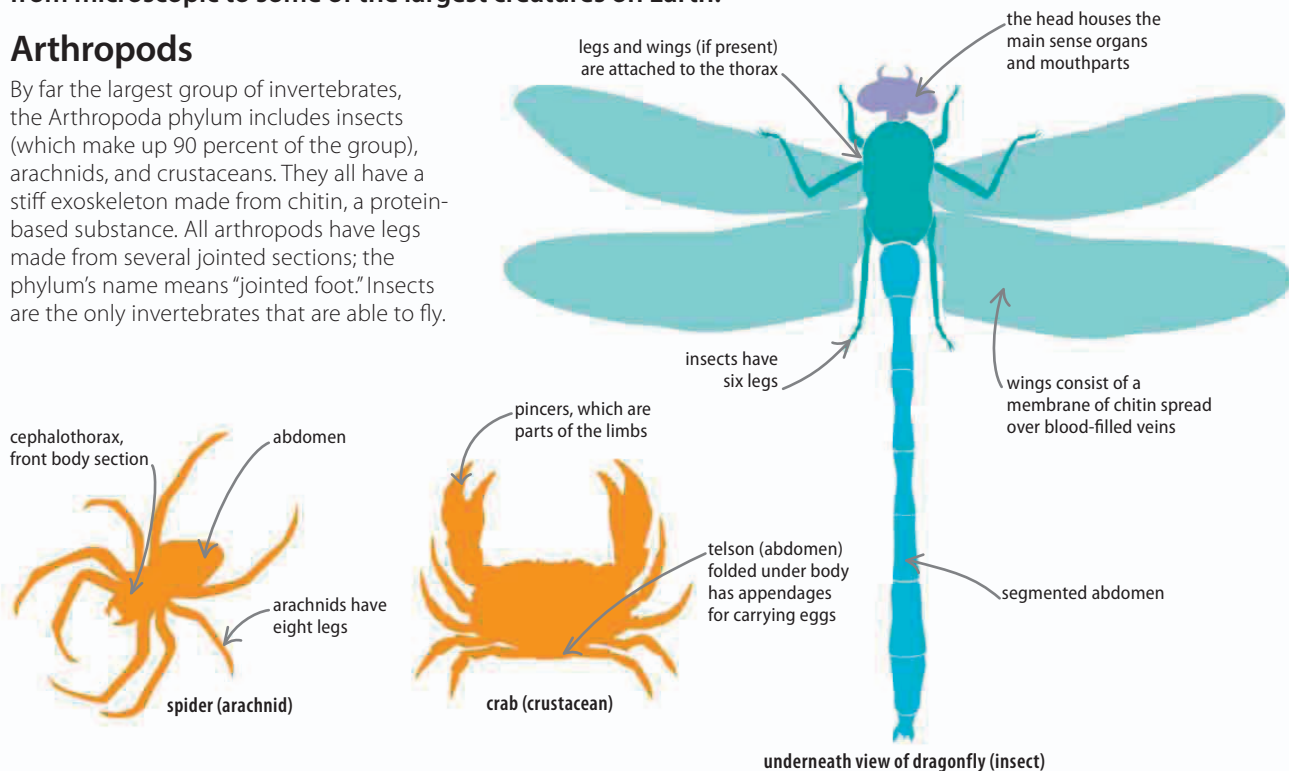
The invertebrates are made up of dozens of phyla, many as distantly related to each other as they are to vertebrate animals. They range from microscopic to some of the largest creatures on Earth.

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- ◀ 39 Anchor points
- ◀ 42 Asexual reproduction
- ◀ 47 Metamorphosis
- ◀ 52–53 Animal relationships

Arthropods

By far the largest group of invertebrates, the Arthropoda phylum includes insects (which make up 90 percent of the group), arachnids, and crustaceans. They all have a stiff exoskeleton made from chitin, a protein-based substance. All arthropods have legs made from several jointed sections; the phylum's name means "jointed foot." Insects are the only invertebrates that are able to fly.

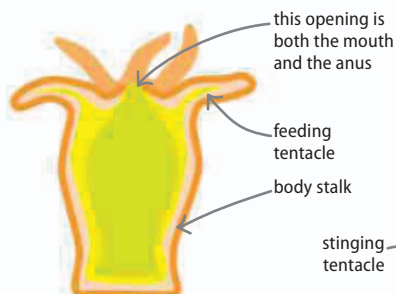


Radiata

Most animals are bilaterally symmetrical, which means they can be divided into two halves that mirror each other. Radiata is a subkingdom (a group of phyla) made up of simple animals with round bodies. Radiata have both radial (symmetry around a fixed point, called the center) and bilateral symmetry. They do not have a mouth as such, but one body opening through which both food and waste pass. The main phylum is the Cnidaria, which includes corals, jellyfish, and anemones. Cnidarians have two types of body form, the polyp and the medusa.

▽ Polyp

The polyp is the upright form used by corals or sea anemones. They sit on the seabed with feeding tentacles facing upward, sifting food from the water.



▽ Medusa

Adult jellyfish are medusae, the bell-shaped form of cnidarians. Medusae are free swimming, and have stinging tentacles that hang down.

